Tillicum Onboarding – IMT 526 A

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INFORMATION TECHNOLOGY
UNIVERSITY of WASHINGTON

Tillicum

LAUNCHED
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2025

University of Washington's next-generation, GPU-accelerated computing platform designed to support cutting-edge research, teaching, and learning in AI, machine learning, data science, and scientific simulation.

WHERE PERFORMANCE MEETS SUSTAINABILITY

RANKED #7 FASTEST US HIGHER ED	RANKED #185 FASTEST WORLDWIDE	RANKED #51 POWER EFFICIENCY

WHY TILLICUM MATTERS?

Broad Academic Access

3	11	33
Campuses	Colleges & Schools	Departments

77	542
Research Groups	Users

Proven Demand & Impact

94,000
GPU Hours Consumed

SUPPORTING ALL DOMAINS

- Engineering
- Math
- Computer Science
- Physical & Life Sciences
- Quantum
- Teaching & Learning
- AI
- Medicine

INFORMATION TECHNOLOGY
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NVIDIA

DELL

[re]DESIGN

USER SUPPORT

INFORMATION TECHNOLOGY
UNIVERSITY of WASHINGTON

eScience Institute

Helpdesk

help@uw.edu

- Hyak, Tillicum, Kopah, Lolo, Cloud
- Research Computing Consulting
- Support Request form

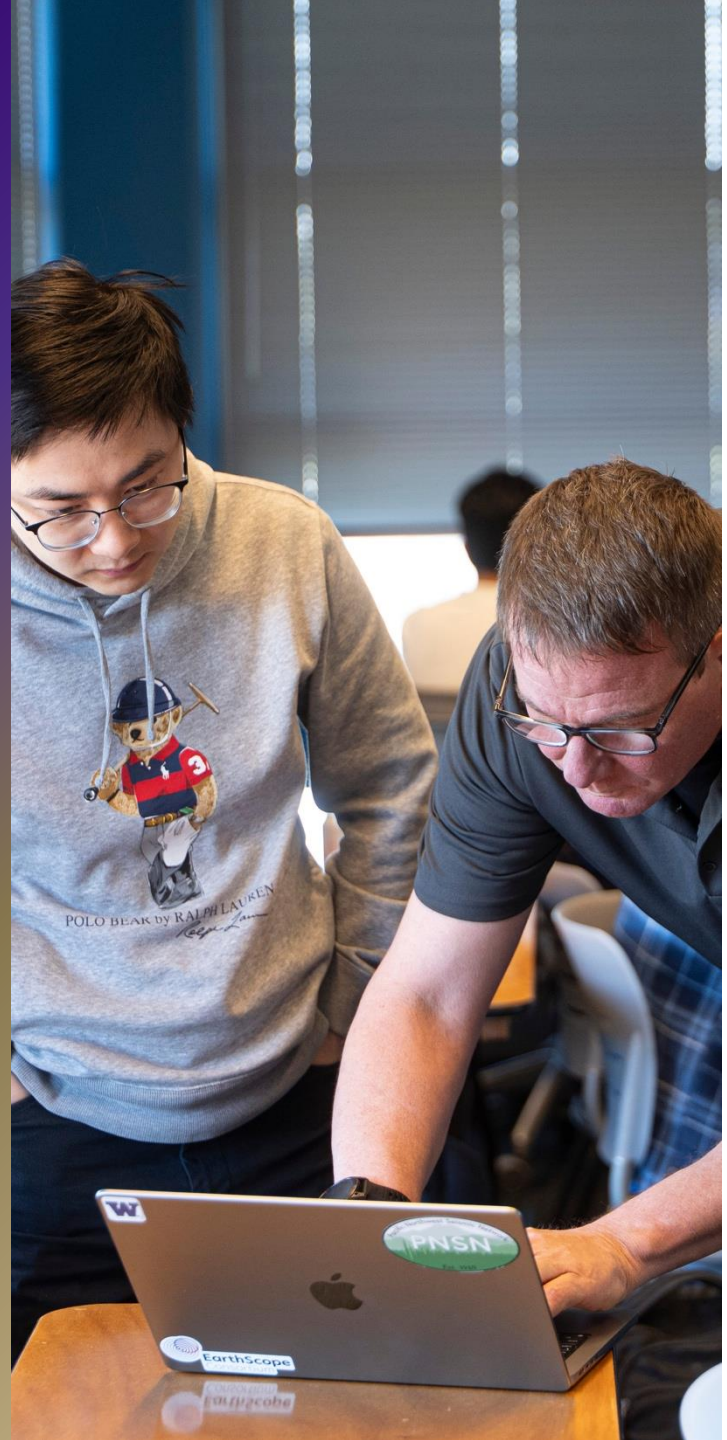
Office Hours

- Drop-in or by Appointment
  - Calendar
- In-person and remote

Regular Training Opportunities

- Recorded
- Events Calendar
- We have a mailing list

Everything underlined is a link



Documentation

www.hyak.uw.edu/

- Step by step guides to tools and software
- Monthly blog posts with updates
- Links to support and services

Learning Resources

www.hyak.uw.edu/learn

- Short How-to videos
- Full length tutorials

Everything underlined is a link

RESEARCH COMPUTING CLUB

STF | Student Technology Fee Committee

Computing Resources

Hyak Klone:

- 26 nodes
- 24 GPUs
- Priority job scheduling
- Computing storage

Tillicum:

- Apply for $250 in Tillicum computing credits

Cloud Credits Program

- Apply for $500 in AWS credits
- Guidance and troubleshooting

Slack Community

- Get help from your peers
- Get notified about events, jobs, and fellowships

Special Events

- Hackathon
- AWS GameDay

Logging in to Tillicum

Log in to Tillicum using SSH:

- Set up two-factor authentication
- Use SSH to log into Tillicum

- Mac and Linux Users: Open the Terminal

- Windows Users: PowerShell / WSL / Git Bash

- Command:

ssh UWNetID@tillicum.hyak.uw.edu

```
 _____ _ _ _ _                                  _    ___
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This login node is meant for interacting with the job scheduler and
transferring data to and from Tillicum. Please work by requesting an
interactive session on (or submitting batch jobs to) compute nodes.

Visit the HYAK website for more documentation:
https://hyak.uw.edu/docs/

Questions? E-mail help@uw.edu with "hyak" in the subject.

Run "scontrol show res" to see any reservations in place that will
prevent your jobs from running with the reason "ReqNodeNotAvail".

Last login: Fri Feb 20 23:20:29 2026 from 98.247.40.157
[kcxie@tillicum-login02 ~]$
```
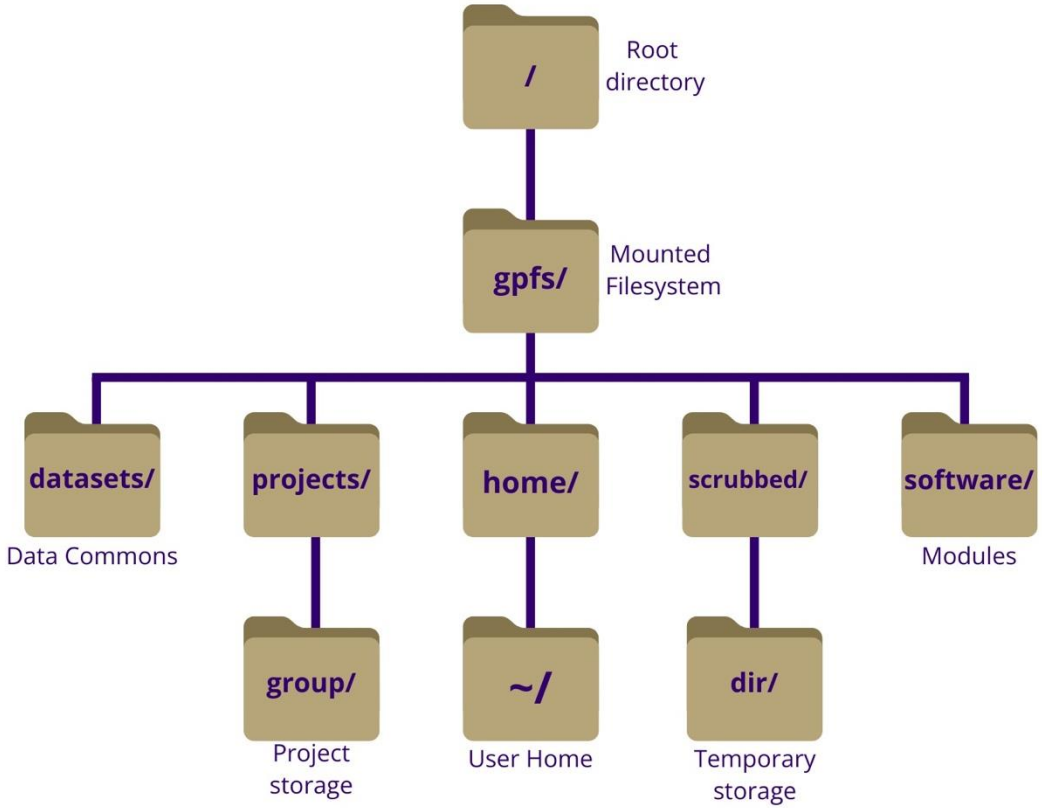

Filesystem and Storage on Tillicum

Storage spaces for Tillicum users:

- Home directory (/gpfs/home/UWNetID): 10GB, personal, backed-up storage
- Project/lab dedicated storage (/gpfs/projects/imt526a): 1TB, shared, backed-up storage for projects
- Scrubbed storage (/gpfs/scrubbed/some-directory): large, temporary scratch space for active computation



Job Scheduling

- Tillicum jobs are submitted under a "Quality-of-Service" or QOS, which defines limits like wall time, GPU count, and concurrent jobs.
- By default, all your interactive sessions on Open OnDemand are submitted to debug QOS for the in-class hands-on sessions.

QOS	Max Time	Max GPUs per Job	Concurrent GPU Limit	Notes
normal	24 hours	16	96 GPUs	Standard production work
debug	1 hour	1	1 job	Quick testing and setup
interactive	8 hours	2	2 jobs	Real-time work or debugging
long	7 days	16	QoS cannot exceed 96 GPUs	Long jobs
wide	24 hours	NA	QoS cannot exceed 96 GPUs	Distributed jobs

Monitoring Your Usage

- Pay-as-you-go by the GPU hours: usage-based billing

- Rate: **$0.90** per GPU-hour

- Formula: ***GPU Hours* = *Elapsed Time* x *N* GPUs**

- `hyakusage`: CLI tool to help users track their usage and costs

Users are responsible for monitoring their own usage.

Monitoring Your Usage

- `hyakusage`: CLI tool to help users track their usage and costs

- Displays usage for all accounts you have access to.

- Shows total GPU-hours and costs per account for current billing cycle.

- Shows active credits (e.g., remaining demo credits) with corresponding expiration dates.

- Integrates user- and account-level budgets (if set) to show progress toward limits.

```
  * Billing cycle starts on the 26th of each month.
  * Usage is counted by job END date.
  * Costs are calculated by base rate ($ 0.9/h) x billable GPU hours,
      where billable GPU hours = raw GPU hours x QOS multiplier.
  * Costs are rounded to nearest cent.

Usage Report for Account account1:
Current Billing Cycle TOTAL Usage: 26.23 GPU hours, $23.61
TOTAL Active Credits: $437.34
  * $347.34 expiring on 2026-02-25
  * $90.00 (no expiration)
```

Usage User Breakdown (2025-12-26 to 2026-01-02)

USER	GPU HOURS (HRS)	JOBS
user1	3.69	16

Usage QOS Breakdown (2025-12-26 to 2026-01-02)

QOS	(MULTIPLIER)	GPU HOURS (HRS)	COST (USD)
interactive	(x 1.0)	3.44	$3.09
normal	(x 1.0)	0.26	$0.23
TOTAL (selected users)		3.69	$3.33

Budget Usage Summary

USER	USED/BUDGET (USD)	USED %	TYPE	ENFORCE
user1	3.69 / 500.00	1%	total	true
account1	23.61 / 1000.00	2%	monthly	false

Conda Module and Custom Environment

Tillicum provides a minimal conda installation that you can utilize to build custom conda environment.

- Conda allows you to create isolated and reproducible environments that include specific versions of Python, libraries, and tools.
- Load the module:

`module load conda`- Initializes conda

- Python:

```
[user@tillicum-login01 ~]$ which python3
/usr/bin/python3
[user@klone-login01 ~]$ python3 --version
Python 3.9.25
```

❌ System default - not the Python we want!

```
[user@tillicum-login01 ~]$ conda activate base
(base) [user@tillicum-login01 ~]$ which python
/gpfs/software/miniforge3/25.3.1-3/bin/python
(base) [user@tillicum-login01 ~]$ python --version
Python 3.12.11
```

✔ Always use Python from a conda environment

Creating a Custom Conda Environment

Customize environment and package locations:

- The default location for custom conda environment is your home directory, which has a storage quota of only 10 GB.
- Configure defaults in $HOME/.condarc

```
$ nano ~/.condarc
envs_dirs:
- /gpfs/projects/imt526a/<class env>/envs
pkgs_dirs:
- /gpfs/projects/imt526a/<class env>/pkgs
always_copy: true
$ module load conda
$ conda create -n myenv python=3.12 numpy scipy pandas matplotlib
$ conda activate myenv
```

Registering Conda Environment as Jupyter Kernel

How to use a conda environment as a IPython Kernel?

- Several ways to accomplish this; we recommend:

```
$ module load conda
$ conda create -n myenv python=3.11
$ conda activate myenv
(myenv) $ conda install ipykernel <other packages>
(myenv) $ python -m ipykernel install --user --name myenv --display-name "Python (myenv)"
```

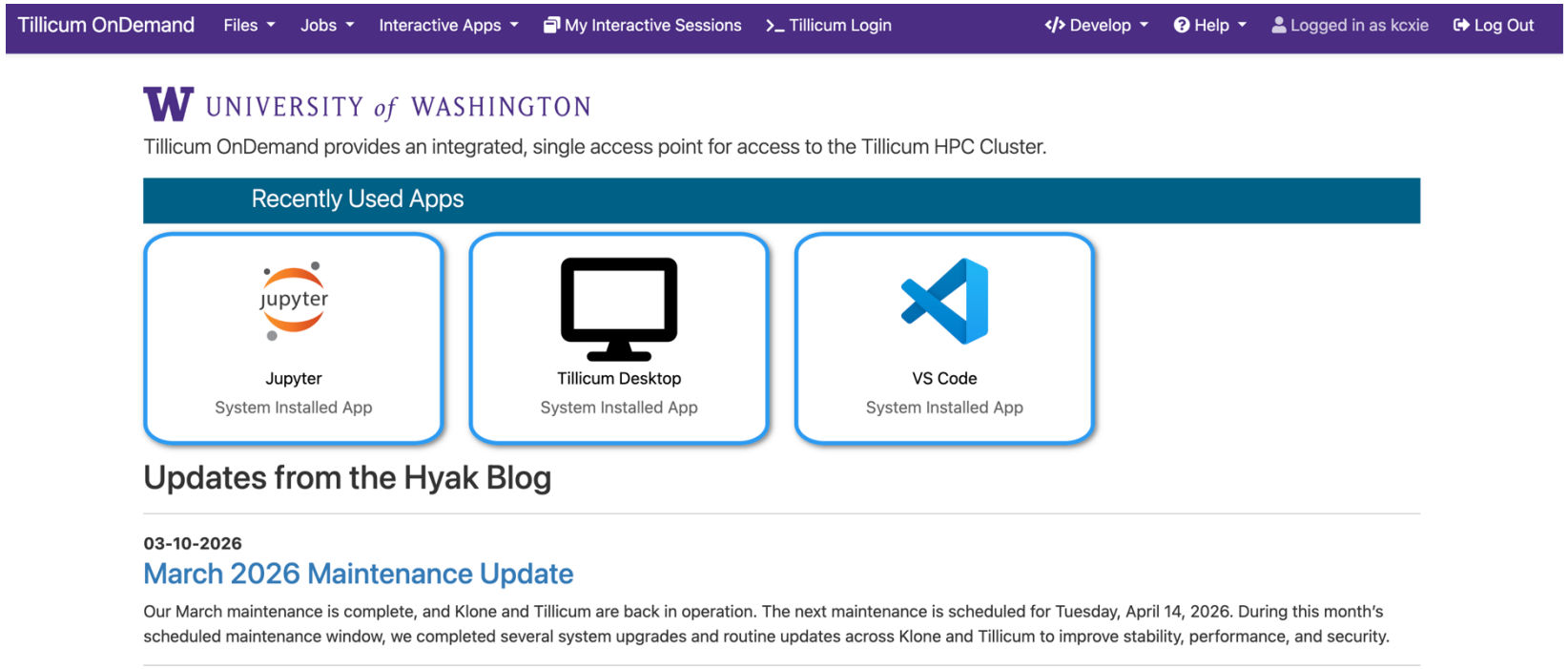
-> This creates a "kernelspec" file: $HOME/.local/share/jupyter/kernels/myenv/kernel.json

- Switch back to your JupyterLab interactive session
- Kernel "Python (myenv)" should be present in the kernel list

How Do I Use Jupyter at Hyak: Open OnDemand

Tillicum: https://tillicum-ood.hyak.uw.edu/

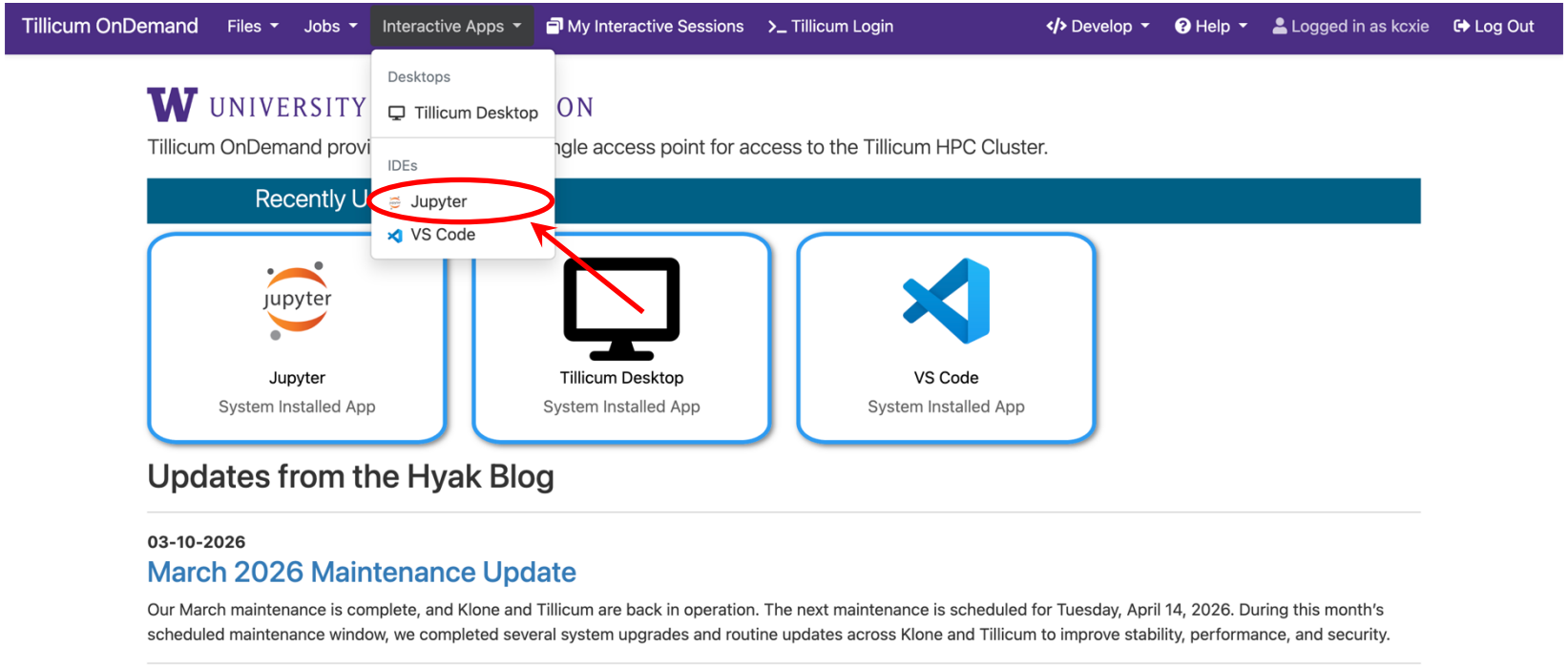
1. Authenticate: If you access OOD from off campus, you will need to connect to the UW VPN. You can find instructions on how to connect to the UW VPN here.



How Do I Use Jupyter at Hyak: Open OnDemand

Tillicum: https://tillicum-ood.hyak.uw.edu/

2. Navigate to the Jupyter App



How Do I Use Jupyter at Hyak: Open OnDemand

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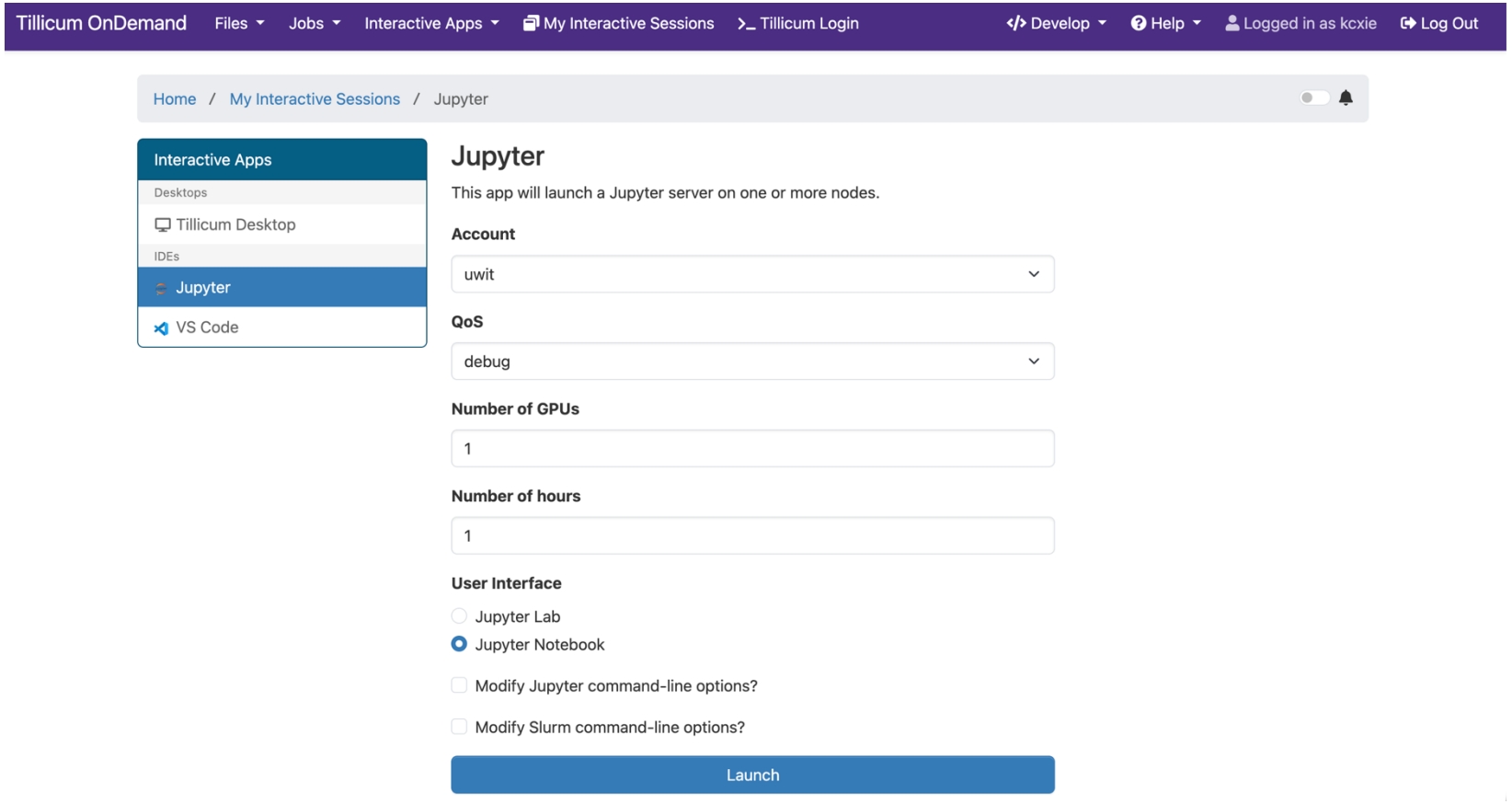
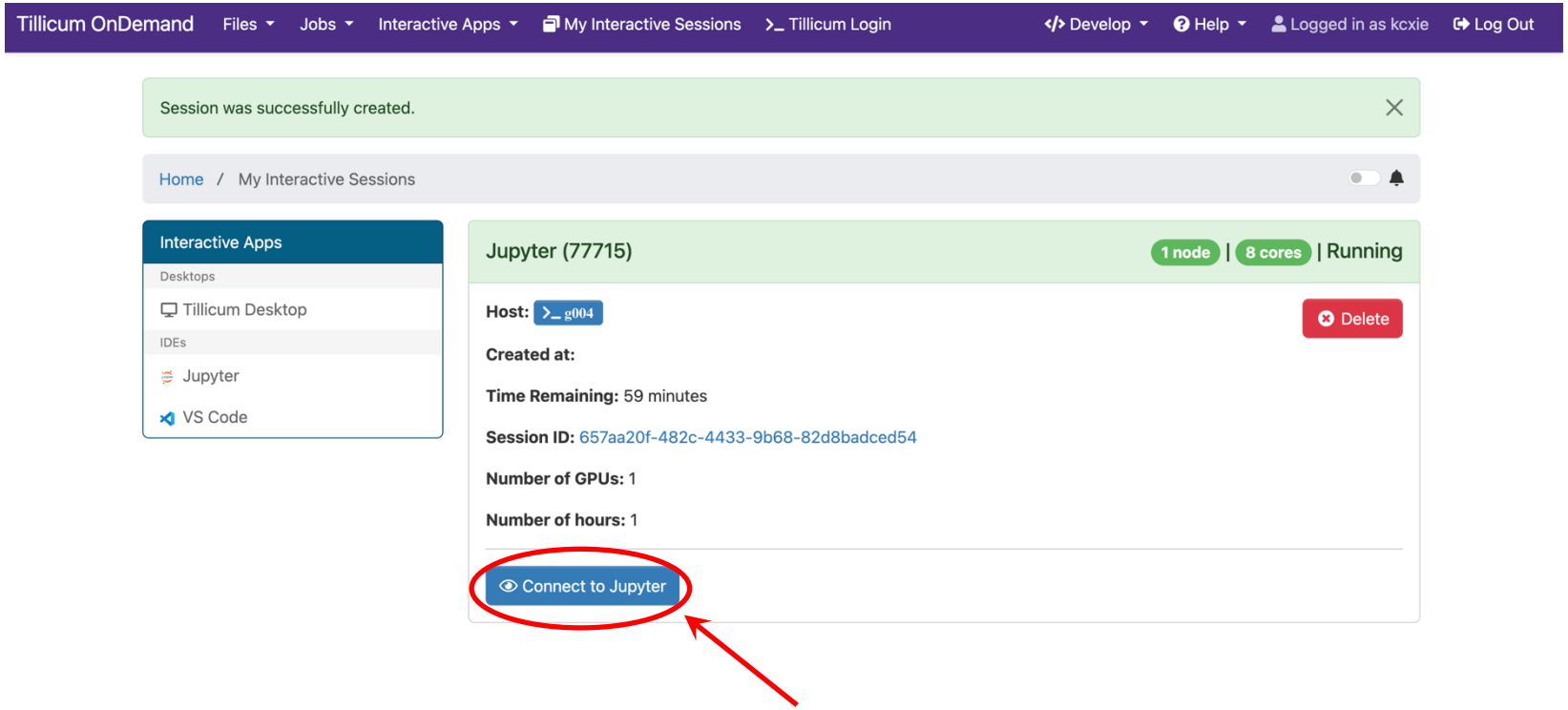
3. Configure the interactive job and launch the job



How Do I Use Jupyter at Hyak: Open OnDemand

Tillicum: https://tillicum-ood.hyak.uw.edu/

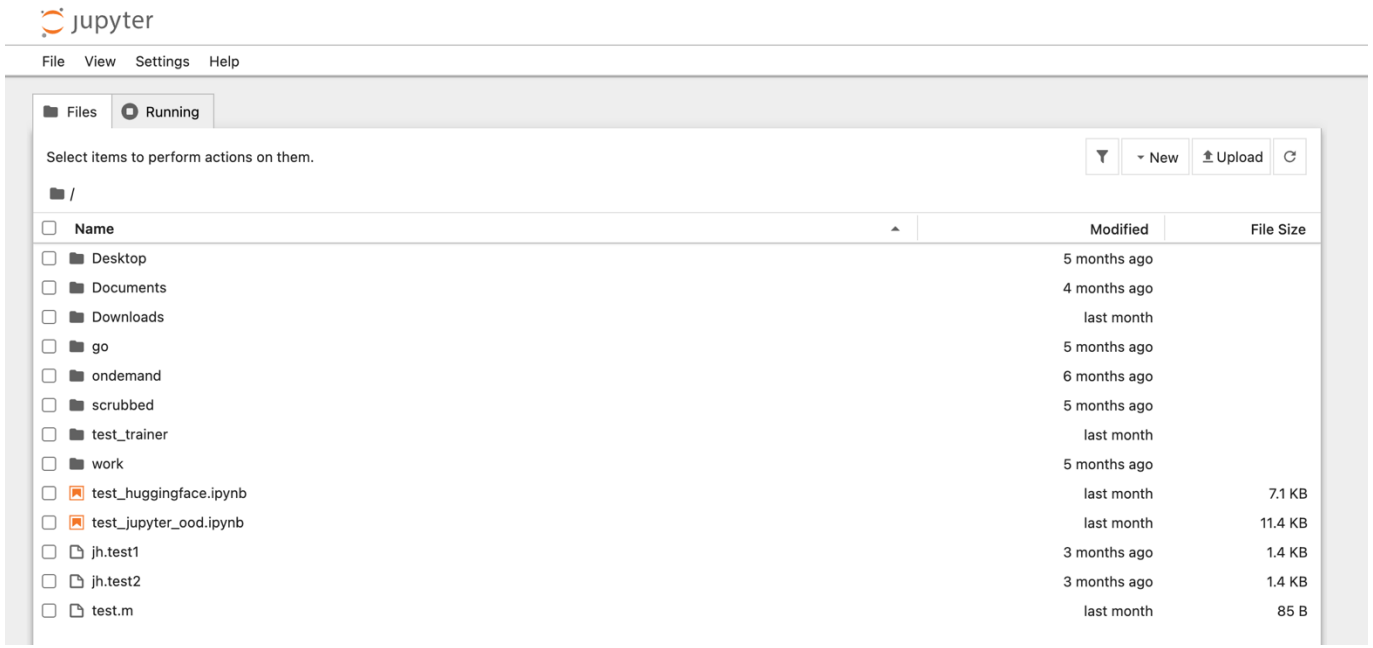
4. Connect to Jupyter



How Do I Use Jupyter at Hyak: Open OnDemand

Tillicum: https://tillicum-ood.hyak.uw.edu/

5. Start a new notebook with a Jupyter kernel

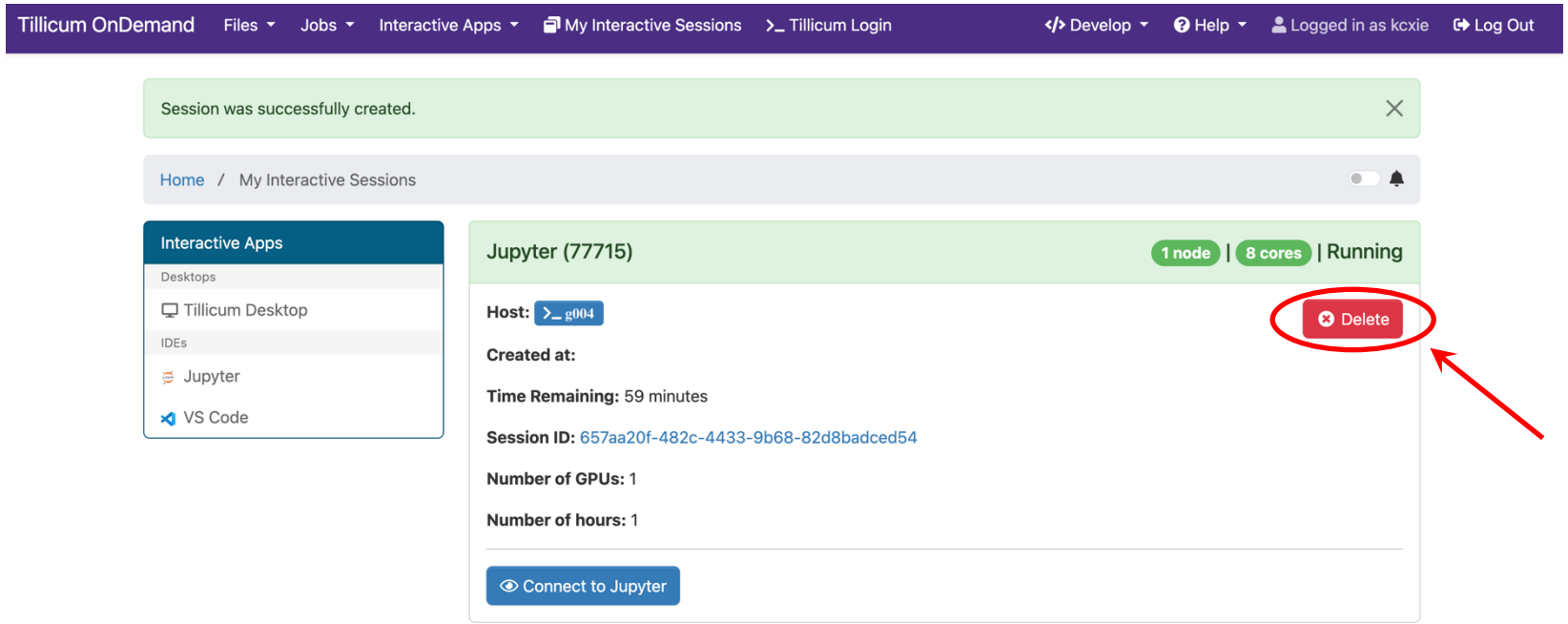


Note: Jupyter is launched from your Home directory.

Closing the Jupyter Session

When you finish:

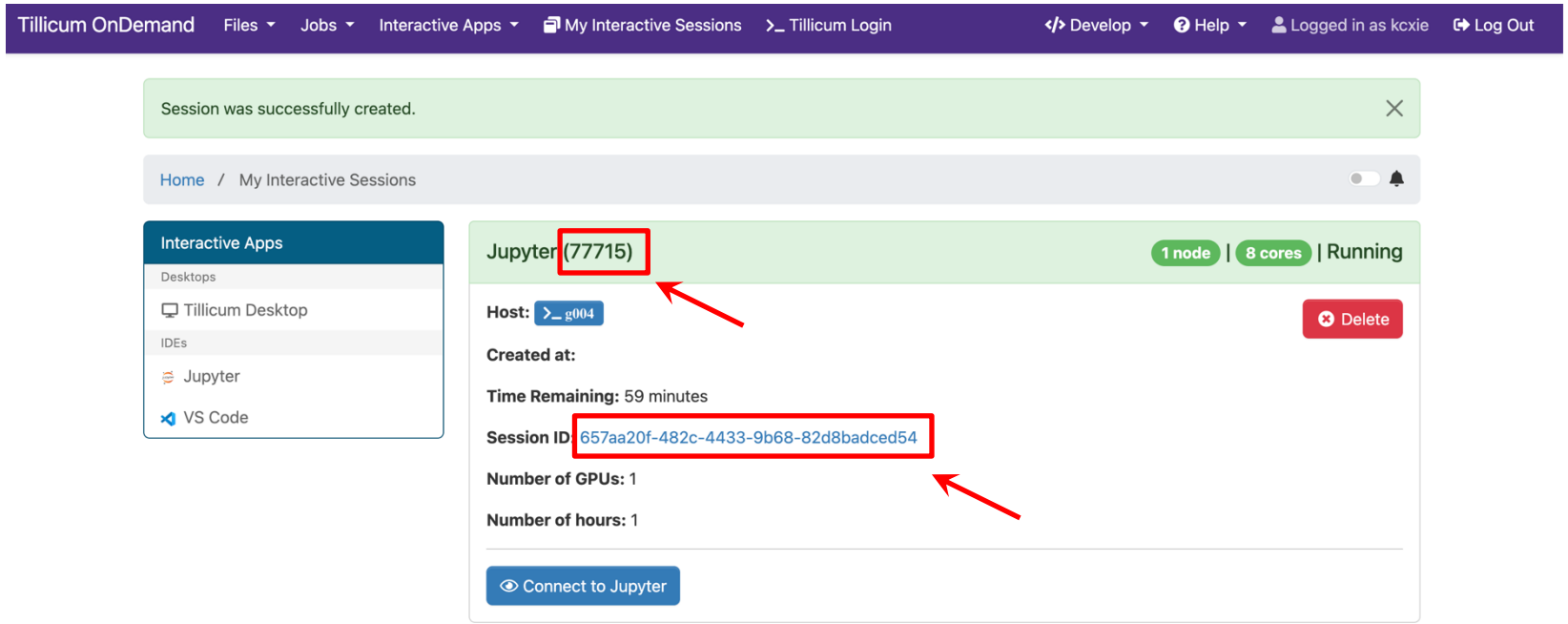
- Save your work and close the Jupyter browser tab.
- Return to OOD My Interactive Sessions and click Delete on your running session card.
- For Tillicum, leaving sessions running consumes GPU hours and counts toward your project usage.



Debugging Jupyter Issues

If you run into any issues, reach out to help@uw.edu with:

- Slurm Job ID — the scheduler identifier (the number in the parentheses).
- Session ID — links to detailed logs for troubleshooting.



Learn more – everything underlined is a link

UWIT - https://it.uw.edu/

Research Computing Services
- Tillicum: GPU-Accelerated Research Computing Platform
  - Intake Form
- Hyak: High-Performance Supercomputing Research Cluster
  - Pricing and Eligibility
  - Documentation
- Data Storage Services
  - Kopah S3 Object Storage
    - Pricing and Eligibility (cost calculator)
    - Documentation
  - Lolo Data Archive
    - Pricing and Eligibility
    - Documentation
- Cloud Computing
- Computing for Restricted Access Data
- Research Computing Consulting

Training and Events
- Hyak mailing list
- Hyak Blog
- UWIT Research Computing Calendar
- Office Hours
- eScience Newsletter (scroll for sign up)
- eScience Data Science and AI Accelerator
- Office of Research Calendar

Past Trainings
- Research Computing Tutorials & Video Library
- eScience YouTube Channel

New Hyak Users
- Free Demonstration Account
- Hyak Basics Tutorial
- Limitations of demonstration accounts
- Using free resources (Checkpoint)

Students
- Research Computing Club
- Student Hyak account
- Cloud Credits
- Student Technology Fee

Other Resources
- **Join the UW AI Community of Practice** on MS Teams to get updates from UW-IT's AI team about events and join the discussion around AI in the news, society, and culture.
- UW Seattle Events Calendar

Special thanks to
- UW Office of Research
- eScience Institute